JavaScript Task's

Sonali Gole(Batch 18 june)

```
// Write a script to swap 2 numbers using third variable.
console.log("Before Swaping");
let num1=Number(prompt("enter first number "));
console.log("Num1: "+num1);
let num2=Number(prompt("enter second number "));
console.log("Num2: "+num2);
num3=num1;
num1=num2;
num2=num3;
console.log("After Swaping");
console.log("Num1: "+num1);
console.log("Num2: "+num2)
```

```
// Write a script to swap 2 numbers without using third variable.
console.log("Before Swaping");
let num1=Number(prompt("enter first number "));
console.log("Num1: "+num1);
let num2=Number(prompt("enter second number "));
console.log("Num2: "+num2);
num1=num1^num2;
num2=num1^num2;
num1=num1^num2;
console.log("After Swaping");
console.log("Num1: "+num1);
console.log("Num2: "+num2)
```

```
// Write a Script to print sum of three number.
let num1=Number(prompt("Enter First number"));
console.log(num1);
let num2=Number(prompt("Enter Second number"));
console.log(num2);
let num3=Number(prompt("Enter Third number"));
console.log(num3);
console.log(num1+num2+num3);
```

```
// Write a script to check eligibility of voting using ternary operator
let age = 19;
age >= 18 ? console.log("You can vote") : console.log("You cannot vote");
```

```
// Write a script to check number is positive or negative using ternary operator.
let num = Number(prompt("Enter number : "));
num > 0 ? console.log("Positive number ") : console.log("Negative Number ");
```

```
// write a script to check variable data and variable datatype are same or not
using ternary operator
let num = 10;
console.log(typeof num);
let n = "Sonali";
console.log(typeof n);
typeof n === String
  ? console.log("value and datatype is same")
  : console.log("Value and datatype is not same");
```

```
// To check a number is even or odd using ternary
let num = 11;
num % 2 == 0 ? console.log("Even number ") : console.log("Odd Number");
```

```
// print maximum of 2 numbers.
let a =4;
let b=6;
a > b ? console.log("a is max "): console.log("b is max");
```

```
// 1. Find whether a number is even or odd.(if-else)
let num = 35;
if (num % 2 == 0) {
  console.log("Even Number");
} else {
  console.log("Odd Number");
}
```

```
//2. Find whether a year is a leap year.
let year = Number(prompt("Enter year"));
if (year % 4 == 0) {
  console.log("Year is leap year");
} else {
  console.log("Year is not a leap year");
}
```

```
// 3. Check Whether the first number is between the second and third
let num1 = Number(prompt("Enter first number :"));
let num2 = Number(prompt("Enter second number :"));
let num3 = Number(prompt("Enter third number :"));
```

```
if (num1 > num2 && num1 < num3) {
  console.log("The first number is between the second and third");
} else {
  console.log("The first number is not  between the second and third");
}
```

```
// 4. Accept amount and apply discount based on the amount range.
let amount = Number(prompt("Enter amount "));
let discount = 0;
let payable = 0;
if (amount >= 10000) {
  discount = amount * 0.3;
  payable = amount - discount;
  console.log("Payable amount is :" + payable);
} else {
  discount = amount * 0.2;
  payable = amount - discount;
  console.log("Payable amount is :" + payable);
}
```

```
// 5. Find the minimum of two  numbers.
let a = 10;
let b = 6;
if (a < b) {
  console.log("a is min");
} else {
  console.log("b is min");
}
```

```
// if-else if-else
// 6. Accept two points (x,y) and determine which quadrant they belong to.
let x = Number(prompt("Enter X "));
let y = Number(prompt("Enter Y "));
if (x > 0 && y > 0) {
  console.log("Point x and y are belongs to quadrant 1");
} else if (x < 0 && y > 0) {
  console.log("Point x and y are belongs to quadrant 2");
} else if (x < 0 && y < 0) {
  console.log("Point x and y are belongs to quadrant 3");
} else {
  console.log("Point x and y are belongs to quadrant 4");
}
```

```
// 7. Accept age,weight and haemoglobin level and check blood donation
eligibility.
let age = Number(prompt("Enter your age :"));
let weight = Number(prompt("Enter your weight :"));
let hb = Number(prompt("Enter your haemoglobin :"));
if (age > 18) {
  console.log("Your age is valid for blood donation");
} else if (weight > 50) {
  console.log("Your weight is valid for blood donation");
} else if (hb > 12) {
  console.log("Your haemoglobin is valid for blood donation");
} else {
  console.log("youvare not eligible for donate blood");
```

```
}
```

```
// 8. Accept month number and print the number of days in month
let month = Number(prompt("Enter month "));
if (
  month == 1 ||
  month == 3 ||
  month == 5 ||
  month == 7 ||
  month == 8 ||
  month == 10 ||
  month == 12
) {
  console.log(`${month} has 31 days`);
} else if (month == 4 || month == 6 || month == 9 || month == 11) {
  console.log(month, `has 30 days`);
} else {
  console.log(month, ` has 28 or 29 days `);
}
```

```
// 9. Accept weekday/weekend and age then calculate price for movie tickit
based on following conditions
console.log("1. Monday");
console.log("2. Tuesday");
console.log("3. Wednesday");
console.log("4. Thursdayday");
console.log("5. Friday");
console.log("6. saturday");
```

```
console.log("7. Sunday");
let day = Number(prompt("Choose day "));
let age = Number(prompt("Enter your age"));
if (day >= 1 && day <= 5) {
  if (age < 13) {
    console.log("Ticket Price is 100");
  } else if (age >= 13 && age < 65) {
    console.log("Ticket Price is 150");
  } else {
    console.log("Ticket Price is 130");
  }
} else {
  if (age < 13) {
    console.log("Ticket Price is 120");
  } else if (age >= 13 && age < 65) {
    console.log("Ticket Price is 180");
  } else {
    console.log("Ticket Price is 150");
  }
}
```

```
// 10. write a program that simulates a traffic light system .
console.log("1. red");
console.log("2. yellow");
console.log("3. green");
let colour = Number(prompt("Choose a colour"));
if (colour == 1) {
```

```
  console.log("Stop");
} else if (colour == 2) {
  console.log("Go Slow");
} else if (colour == 3) {
  console.log("Go");
}
{
  console.log("Invalid input");
}
```

```
// 11. (Switch statement) Check 1)even/odd  2)divisible by 17.
let num = Number(prompt("Enter number "));
let choice = Number(prompt("Enter choice"));
switch (choice) {
  case 1:
    if (num % 2 == 0) {
      console.log(num + " is even");
    } else {
      console.log(num + " is odd");
    }
    break;

  case 2:
    if (num % 17 == 0) {
      console.log(num + " is divisible by 17");
    } else {
      console.log(num + " is not divisible by 17");
```

```
    }
    break;

  default:
    console.log("Invalid operation");
}
```

```
// 12. 1)Leap year Check 2)check positive/negative check

let choice = Number(prompt("Enter choice "));
console.log("1. Check Leap Year ");
console.log("2. Check positive / negative/zero");

switch (choice) {
  case 1:
    let year = Number(prompt("Enter year "));
    if (year % 4 == 0) {
      console.log(year, ` is leap year`);
    } else {
      console.log(year, `is not a leap year`);
    }
    break;
  case 2:
    let num = Number(prompt("Enter your number "));
    if (num > 0) {
      console.log("Number is positive ");
    } else if (num < 0) {
```

```
    console.log("Number is negative ");
  } else {
    console.log("Number is zero");
  }
}
```

//13. 1)maximum of 2 numbers 2)minimum of 2 numbers 3)max from 3 numbers 4) min from 3 numbers

```
console.log("1.Maximum of 2 numbers ");
console.log("2.Minimum of 2 numbers ");
console.log("3.Maximum of 3 numbers ");
console.log("4.Minimum of 3 numbers ");
let choice = Number(prompt("Enter choice"));
let a = Number(prompt("Enter a"));
let b = Number(prompt("Enter b"));
let c = Number(prompt("Enter c"));

switch (choice) {
 case 1:
  if (a > b) {
    console.log(a + " is max");
  } else {
    console.log(b + " is max");
  }
  break;
 case 2:
```

```
  if (a < b) {
    console.log(a + " is min");
  } else {
    console.log(b + " is min");
  }
  break;
case 3:
  if (a > b) {
    if (a > c) {
      console.log(a + " is max");
    }
    console.log(c + " is max ");
  } else {
    if (b > c) {
      console.log(b + " is max");
    }
    console.log(c + " is max");
  }

  break;
case 4:
  if (a < b) {
    if (a < c) {
      console.log(a + " is min");
    } else {
      console.log(c + " is min ");
```

```
      }
    } else {
      if (b < c) {
        console.log(b + " is min ");
      } else {
        console.log(c + " is min ");
      }
    }
    break;
  default:
    console.log(" Invalid operation ");
}
```

```
// 14. 1)area of circle 2) area of triangle 3)area of rectangle

console.log("1. Area of circle ");
console.log("2. area of triangle ");
console.log("3. area of rectangle ");
let choice = Number(prompt("Enter choice"));
switch (choice) {
  case 1:
    let r = Number(prompt("Enter radius"));
    let areaofcircle = Math.PI * r * r;
    console.log("area of circle is " + areaofcircle);
    break;
  case 2:
    let b = Number(prompt("Enter base "));
```

```
    let h = Number(prompt("Enter height "));
    let areaoftriangle = 0.5 * b * h;
    console.log("Area of triangle is " + areaoftriangle);
    break;
  case 3:
    let l = Number(prompt("enter length "));
    let w = Number(prompt("Enter width"));
    let areaofrectangle = l * w;
    console.log("Area of rectangle is " + areaofrectangle);
    break;
}
```

```
// 15. Display Corresponding Day Name(eg. 1-->Monday )

let choice = Number(prompt("Enter choice "));
switch (choice) {
  case 1:
    console.log("Monday");
    break;
  case 2:
    console.log("Tuesday");
    break;
  case 3:
    console.log("Wednesday");
    break;
  case 4:
    console.log("Thursday");
```

```
    break;
  case 5:
    console.log("Friday");
    break;
  case 6:
    console.log("Saturday");
    break;
  case 7:
    console.log("Sunday");
    break;
  case 8:
    console.log("Invalid input ");
}
```

```
// 16. Print all armstrong number upto N.
let N = Number(prompt("Enter Number "));

for (let i = 1; i <= N; i++) {
  let num = i;
  let temp = num;
  let count = 0;
  while (temp > 0) {
    temp = Math.floor(temp / 10);
    count++;
  }
  temp = num;
  let sum = 0;
```

```
  while (temp > 0) {
    let digit = temp % 10;
    sum += Math.pow(digit, count);
    temp = Math.floor(temp / 10);
  }
  if (num == sum) {
    console.log(num + "is armstrong");
  }
}
```

```
// 17. print all palindrom numbers up to N
// Program to print palindrome numbers from 1 to N

let N = Number(prompt("Enter Number: "));

for (let i = 1; i <= N; i++) {
  let num = i;
  let rev = 0;

  while (num > 0) {
    let digit = num % 10;
    rev = rev * 10 + digit;
    num = Math.floor(num / 10);
  }

  if (i === rev) {
    console.log(i + " is Palindrome");
```

```
  }
}
```

```
// 18. Print all disarium numbers upto N.

let N = Number(prompt("Enter Your Number: "));

for (let i = 1; i <= N; i++) {
  let num = i;
  let count = 0;
  let temp = num;

  while (temp > 0) {
    count++;
    temp = Math.floor(temp / 10);
  }

  let sum = 0;
  temp = num;
  let power = count;
  while (temp > 0) {
    let digit = temp % 10;
    sum += Math.pow(digit, power--);
    temp = Math.floor(temp / 10);
  }

  if (sum === num) {
```

```
    console.log(i);
  }
}
```

```
// 19. Print all magic numbers upto N .

let N = Number(prompt("Enter Numbers: "));

for (let i = 1; i <= N; i++) {
  let num = i;
  let sum = 0;

  while (num > 0) {
    let digit = num % 10;
    sum += digit;
    num = Math.floor(num / 10);

    if (num === 0 && sum > 9) {
      num = sum;
      sum = 0;
    }
  }
  if (sum === 1) {
    console.log("Magic Number " + i);
  }
}
```

```
// 20.Print all twin primes upto M.

let N = Number(prompt("Enter Number: "));

for (let i = 1; i <= N; i++) {
    let num1 = i;
    let num2 = i + 2;
    let isTwinPrime = true;



    for (let j = 2; j <= Math.sqrt(num1); j++) {
        if (num1 % j === 0) {
            isTwinPrime = false;
            break;
        }
    }

    for (let j = 2; isTwinPrime && j <= Math.sqrt(num2); j++) {
        if (num2 % j === 0) {
            isTwinPrime = false;
            break;
        }
    }



    if (isTwinPrime) {
```

```
        console.log(num1 + " and " + num2 + " are Twin Prime numbers");
    }
}
```

```
// 21.print numbers from 1 to 100.

let i = 1;
while (i <= 100) {
  console.log(i);
  i++;
}
```

```
// 22. Print numbers from 100 to 5000, incrementing by 50.

let start = 100;
let end = 5000;
while (start <= end) {
  console.log(start);
  start += 50;
}
```

```
// 23. Print numbers from 100 to 1 in descending order.
let i = 100;
while (i >= 1) {
  console.log(i);
  i--;
}
```

```
// 24. print first n even numbers.
```

```
let n = Number(prompt("Enter number"));

let i = 1;
while (i <= n) {
  if (i % 2 == 0) console.log("Even numbers :" + i);
  i++;
}
```

```
// 25. Print welcome message in n times.

let n = Number(prompt("Enter number "));

let i = 1;
while (i <= n) {
  console.log("Welcome");
  i++;
}
```

```
// 26. Print all numbers from 1 to 100.
let start = 1;
let end = 100;
do {
  console.log(start);
  start++;
} while (start <= end);
```

```
// 27.  print numbers from 100 to 5000 , increamenting by 50.
let start = 100;
let end = 5000;
```

```
do {
  console.log(start);
  start += 50;
} while (start <= 5000);
```

// 28.print first n even numbers.

```
let num=Number(prompt("Enter number "))
        let i=1;
        do {
            if (i%2==0) {
             console.log(i);
            }
             i++;

        } while (i<=num);
```

// 29. Accept number n from user and display the sum of odd numbers upto n.

```
let n = Number(prompt("Enter number "));

let i = 1;
let sum = 0;
do {
  if (i % 2 != 0) {
    sum += i;
  }
  i++;
```

```
} while (i <= n);
console.log(sum);
```

```
// 30. Accept a number and display its factors.

let num = Number(prompt("Enter number "));

let i = 1;
do {
  if (num % i == 0) {
    console.log(i);
  }
  i++;
} while (i < num);
```

```
// 31. (Array) Calculate sum and average of elements.

let nums = [18, 29, 31, 42, 53];
let sum = 0;

console.log("Your Array: ", nums);

for (let i = 0; i < nums.length; i++) {
  sum += nums[i];
}

console.log("Sum: " + sum);
console.log("Avg: " + sum / nums.length);
```

```
// 32. find minimum and maximum element in array.

let nums = [10, 20, 34, 56, 89];

console.log("Your Array", nums);
let min = nums[0];
let max = nums[0];
for (let i = 0; i < nums.length; i++) {
  if (nums[i] < min) {
    min = nums[i];
  }
  if (nums[i] > max) {
    max = nums[i];
  }
}
console.log("max :" + max);
console.log("min :" + min);
```

```
// 33. print prime numbers from array.

function isPrime(num) {
  if (num < 2) return false;
  for (let i = 2; i <= Math.sqrt(num); i++) {
    if (num % i === 0) {
      return false;
    }
  }
```

```
  return true;
}

// Accept array input from user
let n = Number(prompt("Enter size of array:"));
let arr = [];

for (let i = 0; i < n; i++) {
  let val = Number(prompt("Enter value:"));
  arr.push(val);
}

console.log("Array: " + arr.join(" "));

// Print prime numbers from array
for (let i = 0; i < n; i++) {
  if (isPrime(arr[i])) {
    console.log("Is Prime: " + arr[i]);
  }
}
```

```
// 34. print palindrome numbers from array.

function isPalindrome(num) {
  let str = num.toString();
  let reversed = str.split("").reverse().join("");
  return str === reversed;
```

```
}

let n = Number(prompt("Enter size of array:"));
let arr = [];

for (let i = 0; i < n; i++) {
  let val = Number(prompt("Enter value:"));
  arr.push(val);
}

console.log("Array: " + arr.join(" "));

for (let i = 0; i < n; i++) {
  if (isPalindrome(arr[i])) {
    console.log("Palindrome: " + arr[i]);
  }
}
```

```
// 35. print even and odd numbers seprately.

let arr = [12, 23, 34, 45, 56, 67, 78, 89];

console.log("Array: " + arr.join(" "));

let evens = [];
let odds = [];
```

```
for (let i = 0; i < arr.length; i++) {
  if (arr[i] % 2 === 0) {
    evens.push(arr[i]);
  } else {
    odds.push(arr[i]);
  }
}

console.log("Even Numbers: " + evens.join(" "));
console.log("Odd Numbers: " + odds.join(" "));
```

```
// 36

let n = 5;
let pattern = "";
for (let i = 1; i <= n; i++) {
  for (let j = 1; j <= n; j++) {
    if (i >= j) {
      pattern += String.fromCharCode(i + 64) + " ";
    } else {
      pattern += "  ";
    }
  }
  pattern += "\n";
}
console.log(pattern);
```

```
// 37
let n = 5;
for (let i = 1; i <= n; i++) {
  for (let j = 1; j <= i; j++) {
    if (i >= j) {
      console.log(i % 2, " ");
    } else {
      console.log(" ");
    }
  }
  console.log();
}
```

```
// 38
let n = 5;
let pattern = "";
for (let i = 1; i <= n; i++) {
  for (let j = 1; j <= i; j++) {
    pattern += "* ";
  }
  pattern += "\n";
}
console.log(pattern);
```

```
// 39.
let n = 5;
let pattern = "";
```

```
for (let i = n; i >= 1; i--) {
  for (let j = 1; j <= i; j++) {
    pattern += "* ";
  }
  pattern += "\n";
}
console.log(pattern);
```

```
// 40.
let n = 5;
let pattern = "";

for (let i = 1; i <= n; i++) {
  // spaces
  for (let j = 1; j <= n - i; j++) {
    pattern += " ";
  }
  // stars
  for (let k = 1; k <= i; k++) {
    pattern += "* ";
  }
  pattern += "\n";
}
console.log(pattern);
```

Task:- create a rgb color system.

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Color Code Generator</title>
    <link rel="stylesheet" href="Task.css" />
  </head>
  <body>
    <div class="container">
      <h1>Color code generator</h1>
      <div id="rgb">rgb(100, 200, 255)</div>
      <div id="colour"></div>
      <button id="generatebtn">Generate Color</button>
      <button id="dark">Change Mode</button>
    </div>
    <script src="Task.js"></script>
  </body>
</html>
```

```css
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
```

```
}
body {
  display: flex;
  justify-content: center;
  align-items: center;
  height: 100vh;
  background-color: #fdfdfd;
}
.container {
  text-align: center;
  border: 2px solid black;
  border-radius: 10px;
  padding: 20px;
  width: 350px;
}
h1 {
  border: 2px solid black;
  display: inline-block;
  padding: 5px 15px;
  border-radius: 10px;
  color: black;
}
#rgb {
  border: 2px solid black;
  display: inline-block;
  padding: 5px 15px;
```

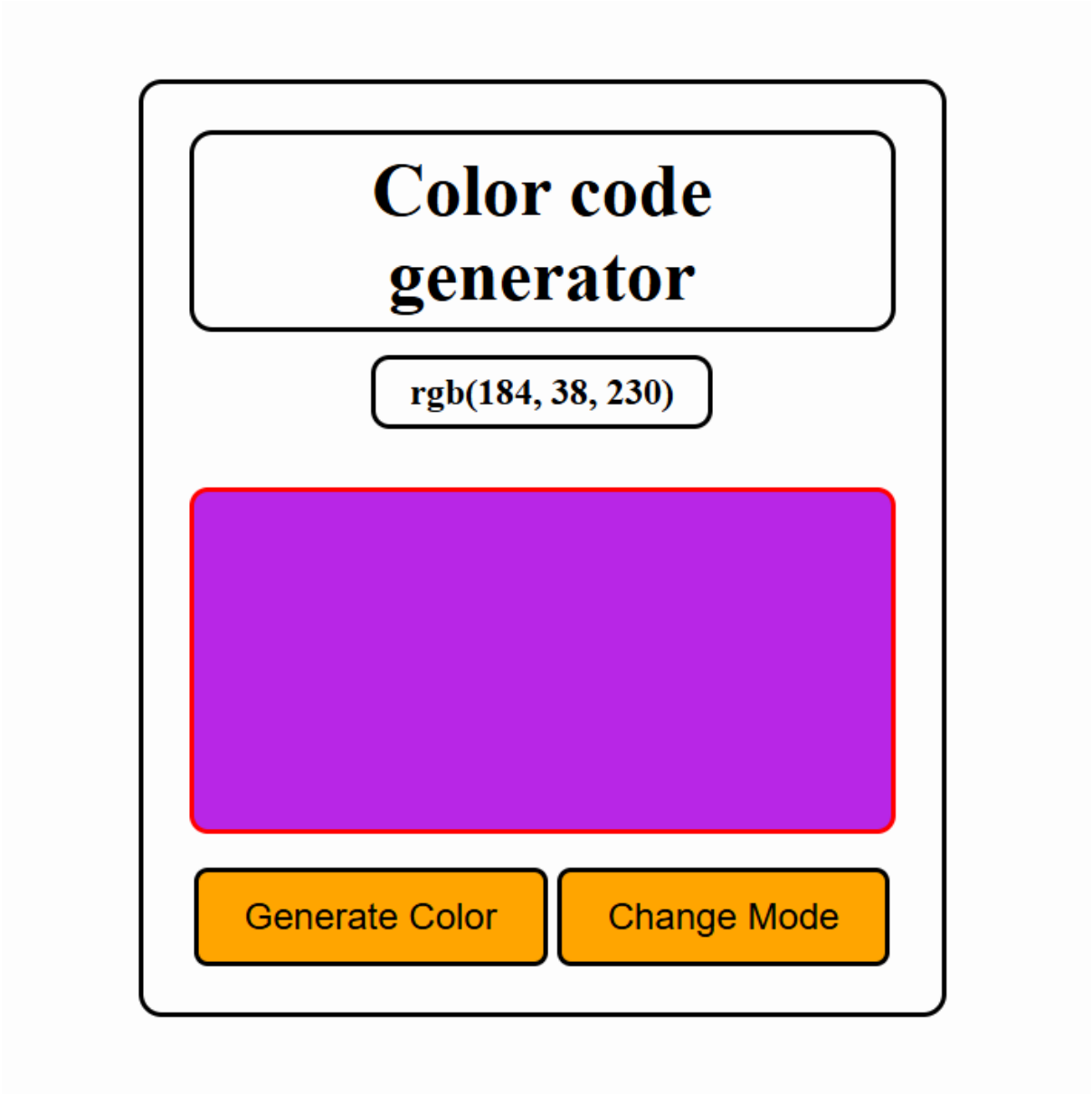
```
  margin: 10px 0;
  font-weight: bold;
  border-radius: 8px;
  color: black;
}
#colour {
  border: 2px solid red;
  height: 150px;
  margin: 15px 0;
  border-radius: 8px;
  color: red;
  line-height: 150px;
  font-weight: bold;
  font-size: 18px;
}
button {
  padding: 10px 20px;
  background: orange;
  border: 2px solid black;
  border-radius: 6px;
  cursor: pointer;
  font-size: 16px;
}
button:hover {
  background: darkorange;
}
```

```
.dark {
  background-color: darkgray;
}



const btn = document.getElementById("generatebtn");
const rgbCodeText = document.getElementById("rgb");
const colorBox = document.getElementById("colour");
function generateColor() {
  let r = Math.floor(Math.random() * 256);
  let g = Math.floor(Math.random() * 256);
  let b = Math.floor(Math.random() * 256);
  let rgbCode = `rgb(${r}, ${g}, ${b})`;
  rgbCodeText.innerText = rgbCode;
  colorBox.style.backgroundColor = rgbCode;
}
btn.addEventListener("click", generateColor);
let newbtn = document.querySelector("#dark");
newbtn.addEventListener("click", function () {
  let con = document.querySelector(".container");
  con.classList.toggle("dark");
});
```

Output:-

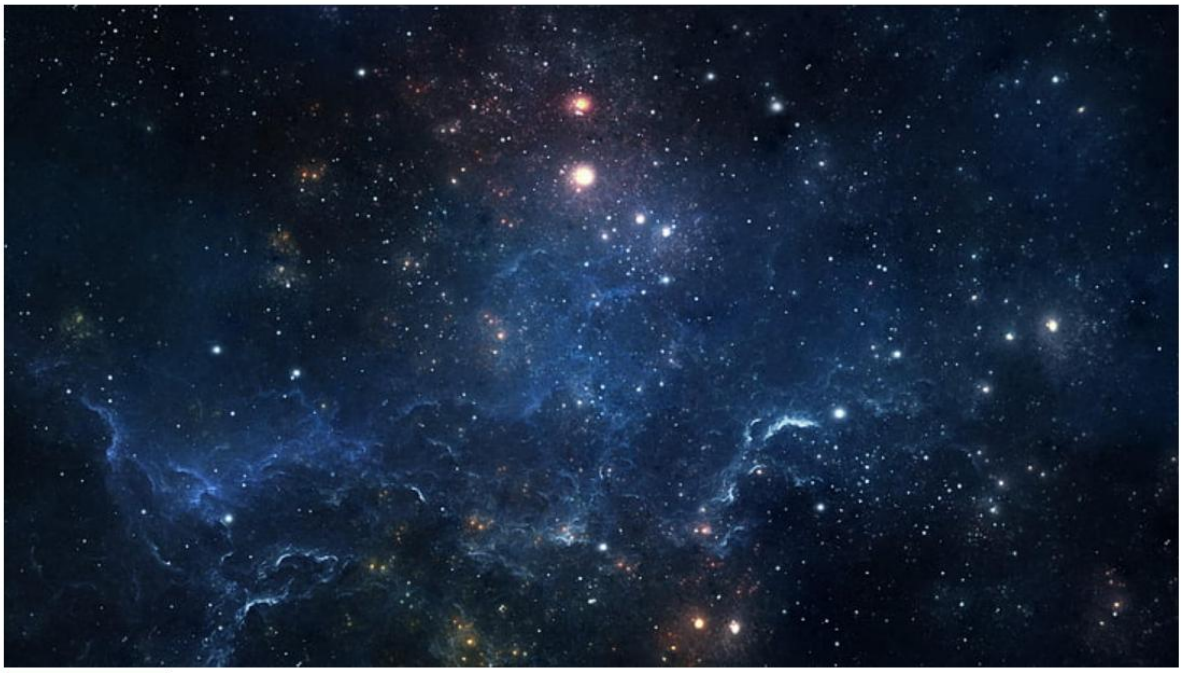


Task:- change image using event listner

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
```

```
    <title>Document</title>
  </head>
  <body>
    <img id="myimg" src="img.jpg" alt="" />
    <!-- <img id="img" src="img1.jpeg" height="200px" alt="" /> -->
    <div>
      <button
        onclick="changeImage()"
        style="height: 30px; background-color: plum"
      >
        change Image
      </button>
    </div>
    <script>
      function changeImage() {
        document.getElementById("myimg").src = "img1.jpg";
      }
    </script>
    <script src="Task1.js"></script>
  </body>
</html>
```

Output:-



change Image

Task:-Form

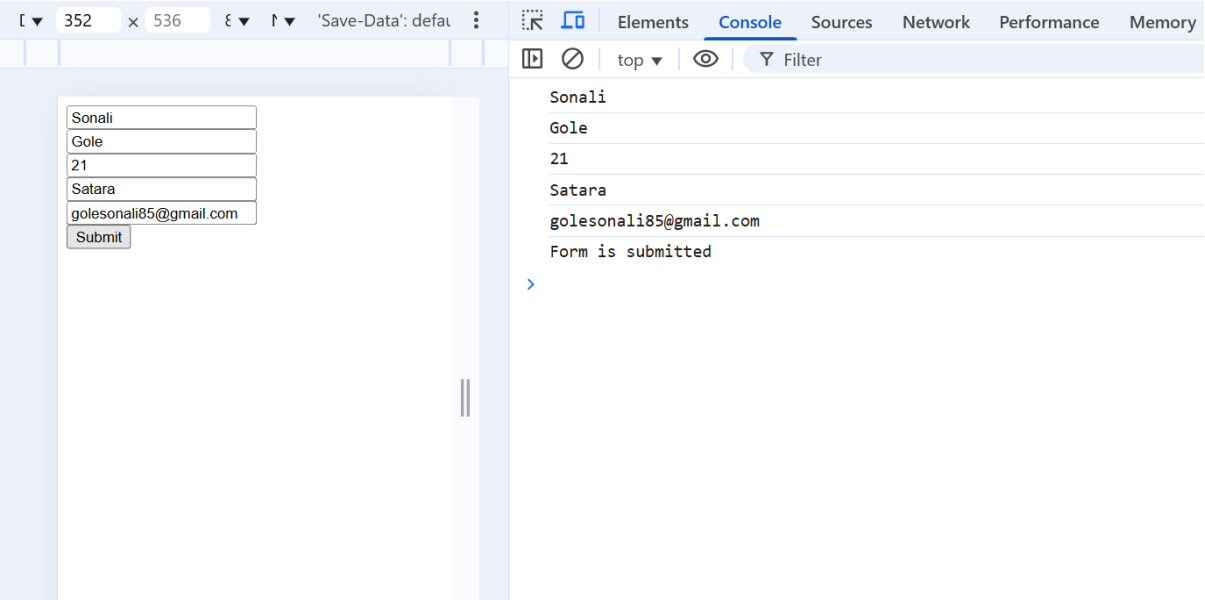
```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Document</title>
  </head>
  <body>
    <form action="/">
      <input type="text" placeholder="Enter first name " /><br />
      <input type="text" placeholder="Enter last name " /><br />
      <input type="number" placeholder="Enter age" /><br />
```

```
    <input type="text" placeholder="Enter city" /><br />
    <input type="email" placeholder="Enter email" /><br />
    <button>Submit</button>
   </form>
   <script src="Form.js"></script>
  </body>
</html>

// Accepting and dipsplaying vaues from form and displaying on console

let form = document.querySelector("form");
form.addEventListener("submit", (event) => {
  event.preventDefault();
  let input = document.querySelectorAll("input");
  Firstname = input[0].value;
  Lastname=input[1].value;
  age = input[2].value;
  city=input[3].value;
  email=input[4].value;
  console.log(Firstname);
  console.log(Lastname);
  console.log(age);
  console.log(city);
  console.log(email);
  console.log("Form is submitted");
});
```

Output:-


352
536
'Save-Data': defau
Sonali
Gole
21
Satara
golesonali85@gmail.com
Submit
Elements
Console
Sources
Network
Performance
Memory
top
Filter
Sonali
Gole
21
Satara
golesonali85@gmail.com
Form is submitted